

HelixVPN MVP — Testing & QA Implementation Strategy

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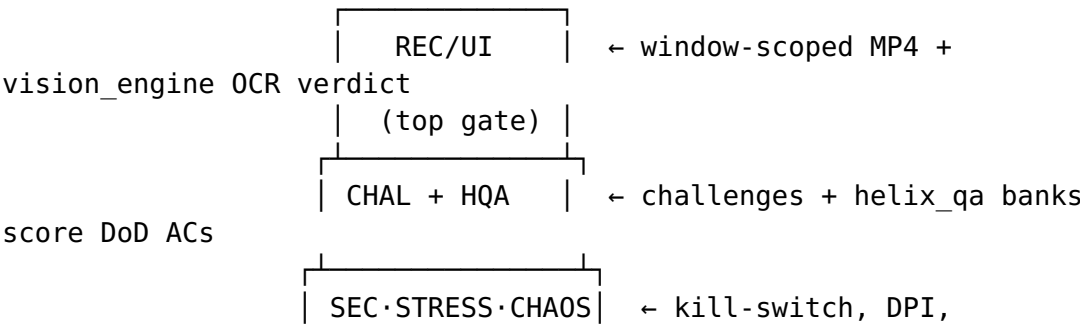
Revision: 1
Last modified: 2026-07-05T14:20:00Z
Status: active — Phase 4 QA architecture deliverable
Authority: docs/research/mvp/final/10-testing-acceptance-and-qa.md (canonical QA contract)
Scope: Full testing strategy, test pyramid, CI/CD integration plan, and links to HelixVPN-specific Challenge / HelixQA banks.

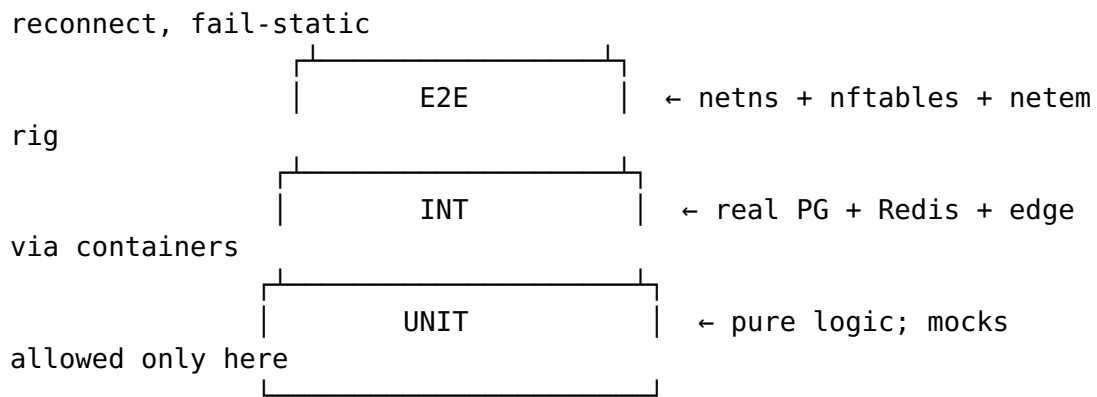
1. One-sentence rule

A green test suite is not proof the feature works. It only counts as proof if a real artifact — a packet capture, a throughput number, a database rowset, a screen recording — was captured *while the feature ran* and shows the user-visible outcome actually happened.

This document operationalizes that rule for HelixVPN.

2. Test pyramid





Layer responsibilities

Layer	Proves	Harness	Mocks?
UNIT	Policy compiler, IPAM, delta diff, MASQUE framing, status FSM	cargo test, go test, dart test	Only here
INT	Enrollment, RLS, reconcile, cert rotation, presence TTL	containers/pkg/boot PG+Redis+edge	No
E2E	Real tunnel carries real packets; authorized reach; default-deny	Netns + nftables + netem rig	No
SEC/STR/CHAOS	Kill-switch, DNS leak, DPI evasion, fail-static, reconnect	rig/, scripts/stress_chaos.sh	No
CHAL/HQA	DoD ACs scored on captured evidence; defeats B1-B5	submodules/challenges/, submodules/helix_qa/	No
REC/UI	User-visible flows rendered and match core FSM	panoptic + vision_engine	No

3. Test-type taxonomy (§11.4.169)

Code	Type	MVP status	Owner subsystem
UNIT	Unit	required	helix-core, helix-go, helix-ui
INT	Integration	required	svc-*, helix-core
E2E	End-to-end	required	netns rig / edge

Code	Type	MVP status	Owner subsystem
FA	Full-automation	required	make spike / make qa
SEC	Security	required	security / helix-core
DDOS	DDoS / load-flood	parked (GAP-6 closed)	svc-api (NFR-413), helix-edge (NFR-414)
STRESS	Stress	required	svc-coordinator, helix-edge
CHAOS	Chaos	required	svc-coordinator, helix-edge
CONC	Concurrency	required	svc-ipam, svc-coordinator
RACE	Race / deadlock	required	helix-core, helix-go
MEM	Memory	required	shim-apple, svc-coordinator
BENCH	Benchmarking	required	bench.sh, criterion
PERF	Performance	required	svc-coordinator, helix-edge
SCALE	Scaling	partial MVP / full P2	svc-coordinator
UI	UI / UX	required	helix-ui
REC	Recorded evidence	required	panoptic
CHAL	Challenges	required (DoD)	submodules/challenges/ helix_vpn/
HQA	HelixQA	required (DoD)	submodules/helix_qa/ banks/helix_vpn/

4. CI/CD integration plan

Constraint: docs/research/mvp/final/10-testing-acceptance-and-qa.md §9 / §11.4.156 — no active remote CI. All gates are local.

Local gate layers

```
pre-commit hook
├─ sh -n on shell scripts
├─ schema-lint (no durable flow table)
├─ no-logging payload lint
├─ mutation-residue scan
└─ UNIT (fast)
```

```
pre-push hook
├─ UNIT (full)
├─ INT (containers boots PG+Redis)
├─ SEC subset (secret audit, RLS fixture)
└─ propagation gates (buf breaking-change lint)
```

```
make test
├─ UNIT
├─ INT
├─ E2E (netns rig)
├─ SEC
├─ STRESS + CHAOS
└─ report to qa-results/
```

```
make qa
├─ CHAL/HQA bank run
├─ REC/UI bank run
└─ coverage-ledger regeneration
```

CI/CD parity (future)

When remote CI is re-enabled, the same local targets are wrapped:

```
# .github/workflows/qa.yml (future)
- run: make test
- run: make qa
- run: scripts/verify-coverage-ledger.sh
```

No new test logic lives in CI; CI only orchestrates the existing local harness.

5. Evidence model

Every feature class maps to a §11.4.69 sink-side evidence class.

Feature class	Evidence class	Artifact
tunnel carries traffic	network_throughput	iperf3 -J + pcap with ≥ 1 inner packet
authorized reachability	network_connectivity	curl 200 + pcap SYN→SYN-ACK
default-deny	network_connectivity (negative)	pcap SYN out, zero SYN-ACK
kill-switch / DNS leak	wifi_link / negative	host pcap: zero plaintext / zero :53

Feature class	Evidence class	Artifact
transport escalation	network_connectivity	StatusReport.transport=="masque-h3" + tshark H3
policy reconcile latency	counter_delta	helix_reconcile_seconds p99 < 1s
revoke latency	counter_delta	revoke→edge-enforcement timing CSV
key-never-leaves	negative	FFI boundary scan + log scan
iOS NE memory	MEM	Instruments .trace / footprint RSS
UI connect flow	REC	window-scoped MP4 + vision verdict

6. HelixVPN-specific banks

Challenges

Path: submodules/challenges/helix_vpn/

Challenge ID	Critical path	Evidence
HVPN-CHAL-AUTH-TUNNEL	Auth + tunnel establishment	pcap + iperf3 JSON
HVPN-CHAL-RECONNECT-ROAMING	Reconnect / roaming	pcap + status log
HVPN-CHAL-KILL-SWITCH	Kill-switch	gap pcap
HVPN-CHAL-DNS-LEAK	DNS leak prevention	DNS pcap + resolvers JSON
HVPN-CHAL-CONTROL-PLANE-HA	Control-plane HA	HA pcap + CP down log
HVPN-CHAL-CLIENT-UI-VISUAL	Client UI visual proof	MP4 + vision verdict

HelixQA

Path: submodules/helix_qa/banks/helix_vpn/

Test-case ID	Critical path	Evidence
HVPN-HQA-AUTH-TUNNEL	Auth + tunnel establishment	pcap + iperf3

Test-case ID	Critical path	Evidence
HVPN-HQA-RECONNECT-ROAMING	Reconnect / roaming	pcap + status log
HVPN-HQA-KILL-SWITCH	Kill-switch	gap pcap
HVPN-HQA-DNS-LEAK	DNS leak prevention	DNS pcap
HVPN-HQA-CONTROL-PLANE-HA	Control-plane HA	HA pcap + log
HVPN-HQA-CLIENT-UI-VISUAL	Client UI visual proof	MP4 + vision verdict
HVPN-HQA-NFR413-API-Rate-Limit	GAP-6: API rate limiting	latency + counter CSV
HVPN-HQA-NFR414-Edge-Flood	GAP-6: edge flood resilience	liveness + legit latency CSV

7. GAP-6 closure

Gap: DDOS test type had no owning FR/NFR.

Fix:

- **NFR-413** — control-plane API rate limiting (token bucket per API key/device) → svc-api owns it.
- **NFR-414** — data-plane edge DDoS/UDP-flood resilience → helix-edge / helix-transport owns it.

At MVP the runnable cells are not-applicable-MVP (single-node-selfhost); the banks are authored-now-parked and mechanically re-arm when `deployment.topology == "multi-tenant-ha"` in Phase 2.

8. Acceptance gates

Phase 0 — spike exit gates

Gate	Bar	Evidence	Test types
G1	plain-UDP $\geq$ 80 % bare link	pcap + iperf3 CSV	E2E, BENCH, CHAL
G2	MASQUE $\geq$ 50 % plain-UDP through DPI block	pcap + tshark + CSV	E2E, SEC, BENCH
G3	iOS NE memory headroom $\geq$ 30 %	Instruments .trace	MEM

Gate	Bar	Evidence	Test types
G4	Go-vs-Rust edge decision	edge_compare.csv	BENCH, PERF, RACE
G5	flutter_rust_bridge FFI	FFI round-trip log + UI recording	UNIT, UI
G6	push-based reconcile	reconcile event log	INT, E2E

Phase 1 — MVP DoD

AC	Criterion	Evidence	Primary challenge
AC1	self-host from zero	terminal recording + healthchecks	HVPN-CHAL-FR901-Init-From-Zero
AC2	enroll connector+client; reach authorized host	E2E netns + pcap	HVPN-CHAL-AUTH-TUNNEL
AC3	deny unauthorized host	negative E2E pcap	HVPN-CHAL-FR201-Default-Deny
AC4	auto-escalate to MASQUE	StatusReport.transport + tshark H3	HVPN-CHAL-FR006-Auto-Ladder
AC5	policy edit < 1 s	helix_reconcile_seconds p99	HVPN-CHAL-FR205-Policy-Reconcile
AC6	revoke < 1 s	revoke→edge timing CSV	HVPN-CHAL-FR107-Revoke-Speed
AC7	kill-switch + DNS leak	host pcap zero leak	HVPN-CHAL-KILL-SWITCH, HVPN-CHAL-DNS-LEAK
AC8	no durable log + 3 apps	schema-lint + UX recordings	HVPN-CHAL-FR801-No-Flow-Table, HVPN-CHAL-CLIENT-UI-VISUAL

9. Risk ordering

The suite runs highest-risk first:

1. kill-switch / DNS leak (AC7)
2. default-deny / RLS (AC3)
3. revoke < 1 s (AC6)
4. key-never-leaves (FR-103)
5. authorized reach (AC2)
6. transport escalation (AC4)
7. policy reconcile (AC5)
8. UI / recorded evidence (AC9)

Only after the irreversible-security floor is GREEN does the rest of the pyramid run.

10. Determinism & mutation

- Every PASS runs N=3 (normal) / N=10 (cycle-validation) against the same artifact MD5 + same rig.
 - Divergent runs are auto-FAIL.
 - Every gate has a paired §1.1 mutation that disables the protection and expects RED.
 - Every analyzer ships with golden-good + golden-bad fixtures.
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11. Links

- Full coverage ledger: [coverage-ledger.md](#)
- Canonical QA contract: [../10-testing-acceptance-and-qa.md](#)
- Challenges bank: [submodules/challenges/helix_vpn/](#)
- HelixQA bank: [submodules/helix_qa/banks/helix_vpn/](#)
- Phase 4 findings report: [docs/reviews/mvp-final/findings/phase4-qa-coverage-report.md](#)